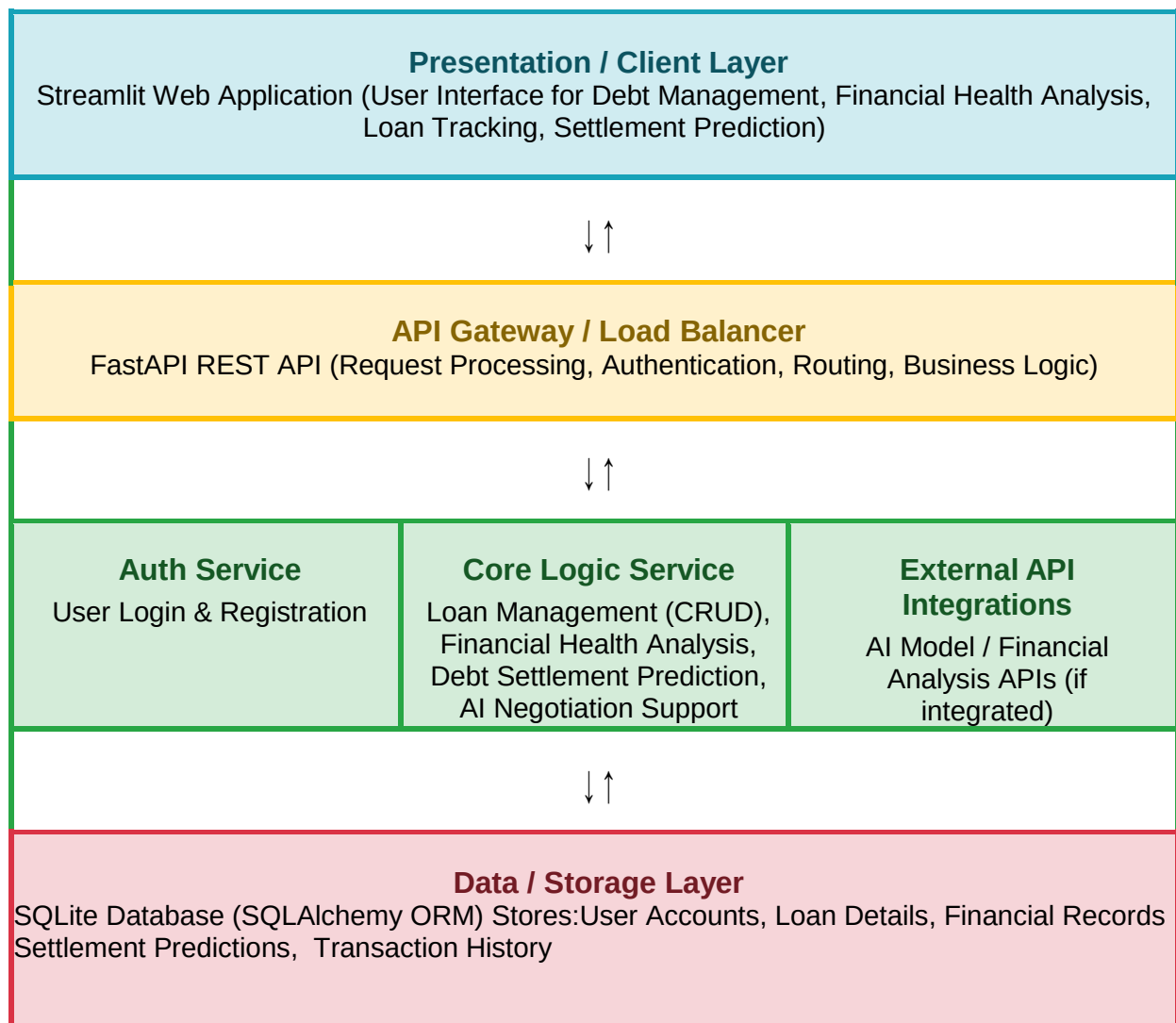


Solution Architecture

Date	28 JUNE 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive dashboard where users can register, log in, manage loans, analyze financial health, and view AI-generated debt settlement recommendations.	Streamlit, HTML, CSS
API Gateway	Receives requests from the frontend, validates inputs, authenticates users, and routes requests to backend services.	FastAPI, Uvicorn
Core Business Logic	Performs loan management, financial health analysis, debt settlement prediction, AI negotiation support, and business rule processing.	Python, FastAPI
Database / Storage	Stores user profiles, loan records, financial data, settlement history, and authentication information securely.	SQLite, SQLAlchemy
External API Integration	Connects the application with AI services or financial APIs to generate intelligent debt relief recommendations.	AI APIs (Optional), Python Requests
Testing Layer	Tests backend APIs, validates business logic, and ensures application reliability.	PyTest